

THE ROCKET

The Geosynchronous Satellite Launch Vehicle (GSLV) Mark-III was conceived primarily to deliver 4,000 kg satellites to geostationary transfer orbit

ISRO has renamed it as LVM-3 to identify its task of placing satellites in a range of orbits, with a carrying capacity of up to 10,000 kg in low earth orbit launches

It is a three-stage rocket with two solid motor strap-ons, a liquid propellant core stage and a cryogenic stage

This was its fifth successful mission, beginning with the first orbital flight in 2017, when it launched India's communications satellite GSAT-19

It was later used to launch Chandrayaan-2 mission to the moon in 2019, which was the rocket's first operational flight

It will also carry Gaganyaan, India's first crewed mission planned in 2024

THE MARKET

India's space economy is expected to grow to $12.8 billion by 2025 (from $9.6 billion in 2020), with launch services segment pegged at $1 billion (up from $600 million)*

There is a growing global demand for launch vehicles to carry out multi-satellite missions, says ISRO chairman S. Somanath

According to him, the Russian rockets are not in consideration anymore, while the commercial potential of Chinese rockets is not accepted by the West

The Ariane 6, which is being developed by ArianeGroup on behalf of the European Space Agency, has also been delayed

All these factors have opened up opportunities for India to enter this niche global satellite launch market, pegged at $6 billion*

OneWeb chairman Sunil Bharti Mittal says they will need about 12-16 launches in the years ahead, with ISRO expected to bag a few of them

**Source: Indian Space Association and Ernst & Young report*